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Fish assemblages composition and structure in three shallow habitats in north Australian tropical bay, Garig Gunak Barlu National Park, Northern Territory, Australia

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The baited remote underwater video technique (BRUVS) was used to compare fish assemblages at three sites with different habitat characteristics: sandy bank (SB), rock reef (RR) and degraded coral reef (DCR) in Port Essington, Garig Gunak Barlu National Park, Northern Territory, Australia. The Carangidae family dominated, representing 35% of all recorded fish. The highest species number was recorded at RR followed by DCR and SB. The highest total fish number was recorded at DCR followed by SB and RR. Fish assemblages from all three sites are clearly different at high confidence level, but still overlapping and higher overlapping was found between SB and RR. Fish assemblages at all three studied sites contained some coral-associated species, fish fauna was less rich compared to coral reefs in the mouth of the bay and fish assemblages were noticeably different from typical coral fish communities. None of the fish assemblages at studied sites were presented by the 'pure', single habitat-associated association of species—at all three sites fish communities were a mixture of fish with various habitat preferences. Reef-associated species have a larger proportion at DCR and at RR their habitat complexity was higher; still soft-bottom habitat fish make substantial fractions in fish assemblages at both DCR and RR. Similarly, fish usually identified as 'reef associated' were found in notable proportions on SB. Fish with wide distribution and low selectivity in habitat preferences comprised a significant part in fish assemblages at all studied sites. In this study the BRUVS technique worked well in the area where diving visual surveys were impossible to implement because of high water turbidity. Another advantage of this method is the non-impact nature of visual survey; they can be used in long-term monitoring of fish assemblages at reference sites.

Keywords: baited underwater video surveys, tropical fish assemblages, fish biodiversity and richness, fish habitats, micro-spatial scale

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INTRODUCTION

Cobourg Marine Park, a part of Garig Gunak Barlu National Park (GGBNP) contains rich and diverse tropical marine fish fauna. Approximately 450 species of fish from about 100 families have been recorded in the Park (Larson & Williams, 1997). Port Essington is a relatively large bay, covering an area of ~240 square kilometres with numerous rocky reefs and sandy banks where many fish are concentrated. Reefs and banks therefore have great importance for amateur fishers visiting the Park. However, there is still a gap in our knowledge of these non-coral reef fish assemblages composition and structure. Worldwide, the great majority of studies have been carried out on coral reefs (Luckhurst & Luckhurst, 1978; Gladfelter *et al.*, 1980; Randall *et al.*, 1997; Sale, 2006). However, only few studies were performed on tropical rocky reefs (Falcón *et al.*, 1996; Arburto-Oropeza & Balart, 2001; Ferreira *et al.*, 2001; Dominici-Arosemena & Wolff, 2006). A number of surveys of fish community composition in northern Australian waters and the Northern Territory were conducted in estuaries and mangrove creeks habitats (Blaber, 1986; Robertson & Duke, 1987, 1990; Blaber *et al.*, 1989, 1990, 1992, 1994; Russell & Houston, 1989; Brewer *et al.*, 1995; Sheaves, 1995; Larson, 1999), in tropical seagrasses (Blaber *et al.*, 1992) and in offshore waters (Russell & Houston, 1989; Blaber *et al.*, 1990, 1994; Brewer *et al.*, 1995). However, information on rocky reef fish assemblages, their composition and structure is absent for the northern tropical Australian mainland.

Presence of large salt water crocodiles and high water turbidity seriously complicate visual diving surveys in Port Essington. Use of baited remote underwater video stations (BRUVS) for fish surveys provides an effective yet non-destructive approach in reef areas, where extractive sampling is unsuitable (Cappo *et al.*, 2003, 2004).

Selecting the appropriate spatial scale for sampling is essential for any fish assemblages study. Recent studies suggest that large-scale (>100 m) habitat features play an important role in structuring reef fish assemblages (Ault & Johnson, 1998; Christensen *et al.*, 2003; Dorenbosch *et al.*, 2004) and it is unclear whether findings from small-scale studies (<3 m^2 plots; Sale *et al.*, 1994; Forrester *et al.*, 2002) can be

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extrapolated to larger-scales (Grober-Dunsmore *et al.*, 2008). Likewise, there is a high possibility of a biased conclusion during an attempt to interpolate data from larger-scale studies to small-scale areas. This is because inshore tropical marine ecosystems have very high spatial habitat heterogeneity and consist of a mosaic of habitat patches with distinct associated fish fauna. Habitat complexity has been positively correlated with fish species diversity and total abundance (Sano *et al.*, 1984; Caley & St John, 1996; Friedlander & Parrish, 1998; Gratwicke & Speight, 2005).

In previous published BRUVS fish surveys (Willis & Babcock, 2000; Willis *et al.*, 2000; Cappo *et al.*, 2004; Watson *et al.*, 2005; Stobart *et al.*, 2007) larger-scale studies have been employed. Several BRUVS devices were either set about 450 m apart along an 1800 m track (Cappo *et al.*, 2004) or 4–7 replicates were made haphazardly within the area of ~1000 m^2 (Willis & Babcock, 2000; Willis *et al.*, 2000) and only one video sample was taken at each randomly selected site. BRUVS apparatus separation served an important objective—to ensure that replicates are independent of one another (Cappo *et al.*, 2004). Such survey design indeed is relevant for unbiased description of fish assemblages of relatively large area, however, it provides scarce information on reef fish assemblages in meso-scale (between 3 m and 100–300 m). Nonetheless, measuring changes in meso-scale (single reef and rock platform) is very important for many purposes, from fish habitat preferences and fish movement studies to environmental monitoring.

In our study a repetitive sampling of the same selected sites was used to find how many video surveys/transects were needed: (a) to describe the greater part of biodiversity of local fish assemblage; and (b) to detect a 25% change in the mean total number of individual fish within the site with sufficient statistical power ≥0.8.

The main aims of this study were to evaluate suitability of BRUVS technique for fish assemblages surveys in shallow tropical habitats with relatively low water clarity. Three sites with different habitat characteristics (sandy bank, degraded coral reef and rocky reef) were selected for this study.

MATERIALS AND METHODS

Sites description and characteristics

This study was carried out in Port Essington, a bay located inside GGNP, Cobourg Peninsula, Arafura Sea, Australia. Three sites with different bottom habitat characteristics were surveyed (Figure 1): (1) a sandy bank with low spatial heterogeneity (Site A) with the area size ~8560 m^2 (11°17.054″ S132°08.811″E). The depth at bank ranges from 4.3 m to 4.9 m (neap tide). Sea floor at this bank, surrounded by deeper waters (depth 12–14 m) consists of silted sand with scarce dead shells; (2) the edge ('reef rock rim') of a degraded coral fringing reef (Site B) (11°16.541″S 132°10.765″ E, area size ~209000 m^2). The depth here ranges from 1.9 m to 4.1 m. The sea floor here consists of rocks and dead massive coral colonies with occasional hydroids and sponges on patches of silted sand; and (3) a rocky reef (Site C) (11°12.252″S 132°06.715″E, area size ~79000 m^2). This site was the deepest (from 3 m to 7 m). Sea bed here consists primarily of rocks with abundant hydroids and sponges, single hard and soft coral colonies and patches of silted sand also covered with epibenthos and epiphytes such as brown, green and coralline algae.

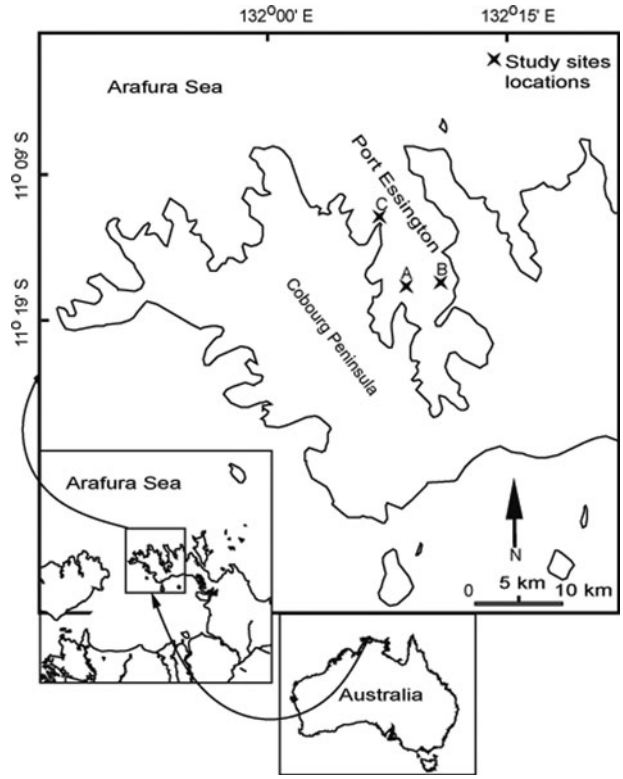

Fig. 1. Geographical locations of baited remote underwater video system (BRUVS) study sites in Port Essington, Garig Gunak Barlu National Park, Cobourg Peninsula, Australia.

At each site, bottom characteristics were assessed by using a series of bottom still images frame-grabbed from video by Adobe Premier 6.0 software. Image Pro Express software was used to obtain per cent area cover of such physical elements of substrate as rocks, boulders, large (>25 cm) coral rubble and macroepibenthos (sponges, soft corals and hydroids) of all areas, contributing to habitat rugosity.

Per cent area cover was calculated using the formula:

$$Pac(r) = (\Sigma S(i..j) * 100)^* S(fw)^{-1}$$

where Pac(r) is per cent area cover, the measure of habitat rugosity, Σ(si..j) – all large objects area cover within the field of the camera view and S(fw) the size of the area of the bottom within the field of the view of the camera. Arcsin transformed per cent cover values for three sites were then compared using one-way ANOVA with data of per cent area cover.

BRUVS description

Two BRUVS were used for all samples, with both consisting of a galvanized roll-bar frame enclosing a simple camera housing with acrylic front and rear ports (Cappo *et al.*, 2004). Sony™ Digital 8 Handicams (model TRV250E) with wide-angle lenses (VCL-0637H) in the housings were used for fish recording. Exposure was set to 'Auto', focus was set to 'infinity/manual', date/time codes were overlaid on footage. 1.5 m bait arms (20 mm plastic conduit) were attached and detached during and after deployment. The bait arm held a 350 mm

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plastic mesh bait canister containing one kilogram of crushed pilchards, *Sardinops neopilchardus* (Cappo *et al.*, 2003, 2004). BRUVS were deployed with 6 mm Spectra ropes and two 30 cm surface floats bearing a flag to make a buoy more visible from the distance.

Survey procedure

All surveys were conducted in March–October 2005 during neap tides to avoid high tidal currents and increased water turbidity during spring tides. Surveys were done in daytime (between 800 hours and 1800 hours). GPSmap76 was used to navigate to each site. The average accuracy of this device is ± 5–7 m. One of the BRUVS apparatus was prepared for video survey (Cappo *et al.*, 2003) and deployed. Depth was measured using the Humminbird 323 sounder. BRUVS remained on the sea floor for 60 minutes, and recorded fish attracted to the bait canister. Site name, date, and the time when survey was started and finished were recorded. Due to the relatively small depth and lack of currents during nip tides, each BRUVS survey point in our study hardly deviates more than ± 14 m from GPS location.

In spite of all attempts to avoid making video surveys in low water transparency about 25% of tapes were unsuitable for analysis. Eventually, nine 60 minute video tapes = replicates were used for analysis of Site A, 15 tapes for Site B and 14 tapes for Site C.

Because of relatively large distances between selected sites (minimal distance 3.67 km between Site A and Site B) and because both BRUVS were never used at the same site, video replicates were independent (e.g. same fish were unable to visit more than one BRUVS during the survey).

Video tapes interrogation and statistical analyses

The area of view 1.5 m away from the video camera lens was 1.7×1.7 m. The depth of area of view depended on visibility underwater. Within this section of Port Essington the highest water transparency at 4–7 m depth rarely exceeds 2 m. Interrogation of video tapes was done according to Cappo *et al.* (2003, 2004, 2007). All 60 minutes of videotape were screened and each new fish species arriving in the field of view of the camera was recorded. Only the maximum number of individuals of each species seen together on the whole tape (MaxN) was used in analyses to avoid fish double-recording during the count. According to previous studies (Priede *et al.*, 1994; Cappo *et al.*, 2004; Watson *et al.*, 2005), MaxN gives a conservative estimate of fish relative density.

Habitat complexity comparisons at different sites were made using one-way ANOVA (statistical package SYSTAT 8.0).

Data on abundance of all recorded species were used to prepare a list of fish for the area of study; to compare assemblages structure according to species habitat preferences and to perform power analysis to assess the number of samples necessary to detect 25% change in total number of individuals within $\alpha = 0.05$ (Harvey *et al.*, 2001). Because rare species will produce underestimates of faunal similarity only data on 31 of the most common species, (Table 1) was used for similarity analyses. PRIMER 6.0 (Clarke & Warwick, 1994) was employed to run DIVERSE, SIMPER and ANOSIM. Four

Table 1. Families, species and fish numbers recorded during BRUVS surveys at three study sites in Port Essington, Cobourg Peninsula (pooled data). Common species are indicated by bold type.

Families species	Number of fish recorded	Rank
Carcharhinidae		
***Carcharhinus melanopterus*, blacktip reef shark**	8	23
***Carcharinhus* sp.**	6	25
Negaprion acutidens, sicklefin lemon shark	1	31
Stegostomidae		
Stegostoma fasciatum, zebra shark	2	30
Gingglymostomatidae		
***Nebris ferrugineus*, tawny nurse shark**	14	18
Hemigaleidae		
Triaenodon obesus, whitetip reef shark	1	31
Dasyatidae		
Dasyatis sp., stingray	1	31
Pastinachus sephen, cowtail stingray	1	31
Taeniura meyeni, blotched fantail ray	5	26
Taeniura sp., fantail ray	2	30
Myliobatidae		
Aetobatus narinari, spotted eagle ray	1	31
Rhinobatidae		
Rhinobatus typus, giant shovelnose ray	2	30
Elopidae		
Elops hawaiensis, giant herring	2	30
Ariidae		
***Arius thallasinus*, giant sea catfish**	7	24
Serranidae		
Cephalopholis boenak, brownbarred rockcod	1	31
***Epinephelus coioides*, goldspotted rockcod**	12	19
Plectropomus maculatus, barcheek coral trout	2	30
Apogonidae		
***Apogon* sp., cardinalfish**	39	10
Carangidae		
***Alectis indicus*, Indian threadfish**	10	22
***Alepes apercna*, smallmouth scad**	34	11
Carangoides chrysophrys, longnose trevally	1	31
***Caranx ignobilis*, giant trevally**	18	17
***Caranx sexfasciatus*, bigeye trevally**	58	6
Caranx tille, Tille trevally	1	31
***Gnathanodon speciosus*, golden trevally**	128	1
***Pantolabus radiatus*, fringefin trevally**	56	7
***Scomberoides commersonnianus*, giant queenfish**	42	8
Scomberoides tol, needleskin queenfish	5	27
***Selar boops*, oxeye scad**	33	12
Trachinotus blochii, snubnose dart	3	29
Ulua aurochs, silvermouth trevally	1	31
Eceneidae		
***Echeneis naucrates*, sharksucker**	11	20
Leiognathidae	69	5
***Leiognathus equulus*, common ponyfish**		
Caesionidae		
Caesio cunning, red-bellied fusilier	3	29
Lutjanidae		
Lutianus argentomaculatus, mangrove jack	3	29
***Lutjanus carponatus*, stripey seaperch**	31	13
***Lutjanus erythropterus*, crimson seaperch**	41	9

Continued

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Table 1. Continued

Families species	Number of fish recorded	Rank
Lutianus gibbus, paddletail	1	31
Lutjanus malabaricus, saddle-tailed seaperch	4	28
Lutjanus russelli, Moses perch	1	31
Nemipteridae		
***Pentapodus porosus*, north-west threadfin bream**	104	2
***Scolopsis monogramma*, rainbow monocle bream**	26	15
***Scolopsis* sp., threadfin bream**	8	23
Haemulidae		
Plectorhinchus gibbosus, brown sweetlips	5	27
***Plectorhinchus labiosum*, painted sweetlips**	19	16
Lethrinidae		
***Lethrinus laticaudis*, grass emperor**	34	11
***Lethrinus nebulosus*, spangled emperor**	7	24
***Lethrinus ornatus*, ornate emperor**	72	4
Lethrinus sp., emperor	4	28
Mullidae		
Upeneus vittatus, striped goatfish	2	30
Ephippidae		
***Platax teira*, roundface batfish**	26	15
Chaetodontidae		
***Chelmon marginalis*, margined coralfish**	11	21
Heniochus acuminatus, longfin bannerfish	5	27
***Parachaetodon ocellatus*, ocellate butterflyfish**	7	24
Pomacanthidae		
Chaetodontoplus duboulayi, scribbled angelfish	2	30
Sphyraenidae		
***Sphyraena jello*, pickhandle barracuda**	30	14
Sphyraena obtusata, striped barracuda	2	30
Labridae		
***Choerodon cyanodus*, blue tuskfish**	7	24
Acanthuridae		
***Acanthurus grammoptilus*, inshore surgeonfish**	79	3
Monocanthidae		
Monacanthus chinensis, fanbelly leatherjacket	2	30
Tetraodontidae		
Feroxodon multistriatus, ferocious puffer	1	31
Lagocephalus lunaris, rough golden toadfish	1	31
Total species recorded: 64	Total individuals: 1115	

root transformations were used when it was necessary to normalize the data.

Jaccard's coefficient of similarity (Sj) compared faunas between Site A and Site B, Site A and Site C and between Site B and Site C by comparing censuses from each pair of habitats:

$$Sj = c * (a + b + c)^{-1}$$

where a and b are the number of species present only in habitats A and B, respectively; and c is the number of species that are common to both habitats (Ludwig & Reynolds, 1988).

Fish assemblages composition derived from this study was then compared with extensive published data for estuarine areas, mangrove creeks, tropical seagrass beds and offshore areas in north Australian waters. Additional information was found in fish taxonomic group descriptions (Allen & Swainston, 1988; Russell, 1990; Allen, 1997) and FishBase (Froese & Pauly, 2002).

RESULTS

Environment characteristics of studied sites

One-way ANOVA found significant differences between the depth at three studied sites (df = 2, F = 13.869, $P < 0.001$). Site B was shallower compared to Site A and Site C. The differences between depth at Sites A and C were not significant ($P = 0.541$, Tukey post-hoc test; Figure 2A). Value of standard deviation is much lower at Site A that reflects flat, low bottom profile at sandy bank.

Differences in habitat complexity between three sites were statistically significant (one-way ANOVA, df = 2, F = 14.84, $P < 0.001$). No significant differences in bottom rugosity was found between Sites B and C while differences between Sites A and B and Sites A and C were highly significant ($P < 0.001$, Tukey test) (Figure 2B). Sites B and C had similar per cent area cover values Pac(r) (63.6% ± 6.9 and 59.3% ± 10.3, respectively). Site A has the lowest habitat rugosity, Pac(r) = 2.8% ± 1.6 (Figure 2B). Videotapes and still images of the sea bottom here show bare silted sand with scarce small fragments of dead shells.

Fish biodiversity and assemblages composition

A total of 64 fish species from 28 families were recorded during BRUVS surveys in Port Essington. The Carangidae family dominated, representing 35% of all fish recorded, while Nemipteridae represented 12.4%, Lethrinidae 10.5% Acanthuridae 7.1% and Leiognathidae 6.2% of all fish recorded. The remaining 23 families represent 28.8% of all fish (Table 1).

Plots at Figure 3A display cumulative numbers of different species observed as each new sample is added. It took 8–9 samples for cumulative curves to approach asymptote, indicating that most of the common species are described.

According to our power calculations, video samples were required for each site to reach statistical power of 0.8. After nine samples, statistical power was close to 1.0 (Figure 3B).

Fish assemblages univariate biodiversity indices

Site C, sandy bank has the lowest recorded fish species number compared to Site B, the degraded coral reef and Site C, rocky reef (Table 2). However, no statistically significant differences were found between study sites in mean species number recorded in any one survey (one-way ANOVA, df = 2, F = 2.393, $P = 0.106$). The highest total fish number was recorded at Site B, followed by Sites A and B. At Site B the mean fish number recorded in one survey was significantly higher compared to Site C (one-way ANOVA, df = 2,

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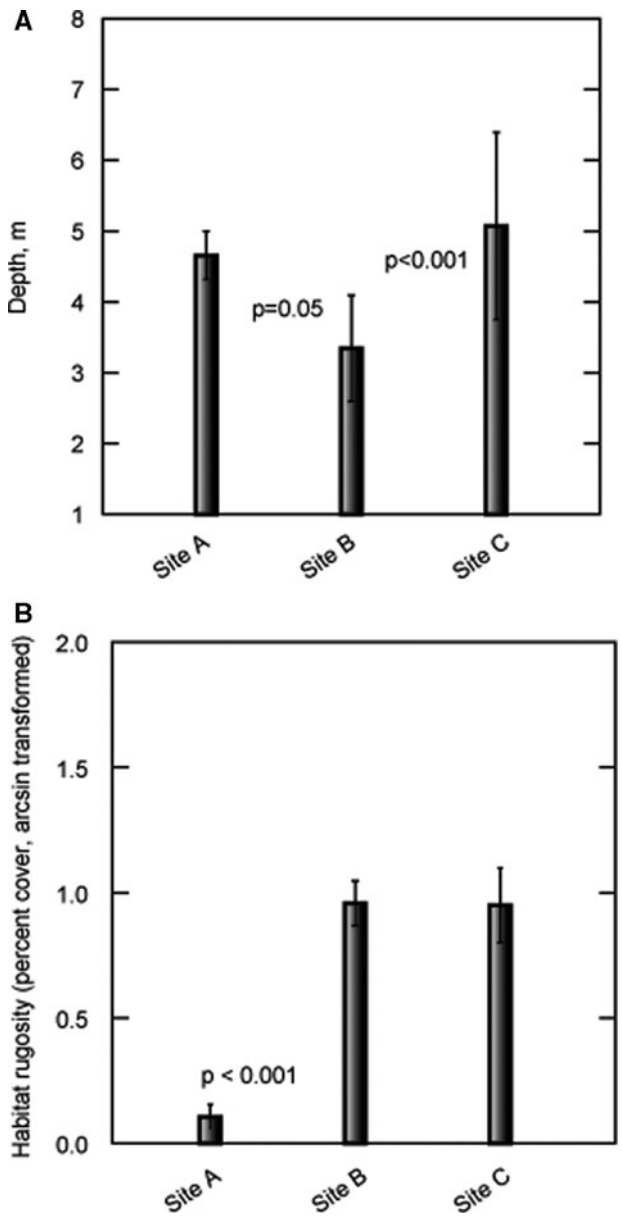

Fig. 2. (A) Mean depth values at three BRUVS survey sites at Port Essington, Cobourg Peninsula. Error bars are standard deviations. Significance level (one-way ANOVA, Tukey HSD test) is given for significantly different values; (B) bottom habitat rugosity expressed as per cent bottom cover (arcsin transformed) of physical structures (rocks, boulders and coral rubble) and macroepibenthos at three studied sites at Port Essington, Cobourg Peninsula. Error bars are standard errors. Significance level (Tukey post-hoc MSD test) is given for statistically different values. Vertical bars are standard errors.

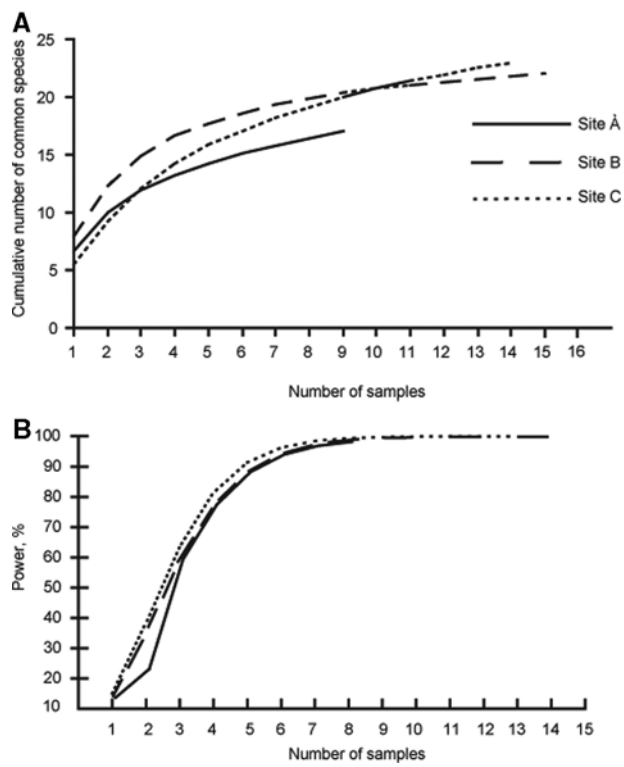

Fig. 3. (A) Plots of the cumulative numbers of the most common species recorded as each new sample is added at three survey sites at Port Essington; (B) power analysis calculation for three BRUVS survey sites at Port Essington.

F = 4.024, $P = 0.027$) while no significant differences were found between Sites A and B (Table 2).

Comparison did not show significant differences in values of all employed standard species richness and evenness indices (Table 2). Values of Pielou's evenness index (J′) were close to 1, and values of modified Hill's ratio (N) did not differ significantly among sites. These indicate a tendency to equal abundance of all species or high evenness within a set of samples (Routledge, 1980).

Jaccard's coefficient of similarity values indicated that relatively few species were in common between sandy bank and dead coral reef (Cj = 0.10 ± 0.02), sandy bank and rocky reef (Cj = 0.20 ± 0.06) and dead coral reef and rocky reef (Cj = 0.15 ± 0.01) comparison when only the most common species were considered. Differences in coefficient values were not significant (one way ANOVA df = 2, F = 2.289, $P = 0.123$).

The sample statistic (Global R) value in ANOSIM analysis was 0.401 at high significance level ($P = 0.001$). This indicates that the probability of the three groupings arising by chance was extremely low and therefore H_o (no differences in fish assemblages in the three analysed Sites A, B and C) can be rejected. Statistic has the highest value in the Site A–Site B pair comparison (R = 0.624, $P = 0.001$), a medium value in Site B–Site C test (R = 0.317, $P = 0.001$). The lowest value of R statistic was in Site A–Site C comparison (R = 0.28, $P = 0.005$) indicating broader overlapping between these assemblages. Analysis shows that fish assemblages from Sites A, B and C are overlapping, but still clearly different at high confidence level.

High values of Bray–Curtis dissimilarity coefficient in SIMPER analysis show substantial differences in compared fish assemblages. The highest coefficient values were found in sandy bank and degraded coral reef comparison—88.78%. Seventeen fish contributed to 90.78% of cumulative assemblages dissimilarity. Some of the species present at one site were absent at another site (Table 3). The Bray–Curtis coefficient in the degraded coral reef and rocky reef comparison was 86.14% with 19 species contributing to 91.72% of cumulative dissimilarity. Average dissimilarity value between Site A and Site C was the lowest (82.80%) with 15 species listed as major contributors to 90.32% of assemblages dissimilarity (Table 3).

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Table 2. Univariate biodiversity indices for fish assemblages at three studied sites at Port Essington.

Location and number of surveys	Total number of recorded species, N	Total number of fish recorded, N	[1]Mean species number in one survey	P	Mean fish number in one survey, N	P	Species richness Margalef, d	P	Brillouin index B1	P
Site A, 9	28	283	8.33 ± 1.18	0.11	31.44 ± 4.74		2.14	0.3	1.48	0.12
Site B, 15	38	554	10.07 ± 0.91	ns	34.64 ± 4.98*	0.022*	2.58	4	1.40	ns
Site C, 14	42	278	7.21 ± 0.94		19.79 ± 3.48*		2.13	ns	1.18	
Hill's number of abundant species in the sample N1	**P**	**Hill's number of very abundant species in the sample, N2**	**P**	**Shannon index (log e base), H′**	**P**	**Pielou's evenness, J′**	**P**	**Modified Hill's ratio, N**		**P**
6.15	0.52	5.05	0.70	1.79	0.41	0.86	0.78	6.51		0.51
6.20	ns	4.68	ns	1.73	ns	0.17 ns		6.22		ns
5.27		4.32		1.56		0.83		5.27		

[1], mean ± SE; ns, differences are statistically not significant; *, indicate significantly different values.

Again, similarly to ANOSIM analysis, the lowest average dissimilarity was found between sandy bank Site A and rocky reef Site C fish assemblages.

The non-metric multi-dimensional scaling (MDS) showed a clear separation between samples from all sites (Figure 4). Stress level 0.15 indicates a potentially useful 2-dimensional ordination. Still, a fine structure of groups is slightly different: Sites A and B samplings have compact groups, while samples from Site C, a rocky reef have very loose grouping. Rocky reef samples occupy intermediate position and overlap both sandy bank samples and degraded coral reef samples.

DISCUSSION

Variability in depth is a crude index of bottom relief (Núñez-Lara & Arias-González, 1998). Higher depth variability at Sites B and C, compared to Site A (Figure 2A) is consistent with higher bottom habitat complexity at former sites (Figure 2B). The lowest number of species recorded at Site A (Table 2) supports the idea that habitats with high structural complexity typically sustain more species than nearby less complex habitats (Bell & Galzin, 1984) because fish use habitats with high complexity as a refuge from predation (Hixon & Beets, 1993; Caley & St John, 1996).

The number of fish taxa recorded at studied sites during the study (Table 1) is approximately half of the number of fish taxa that can be found within a small size coral reef located at the mouth of Port Essington (Gomelyuk, 2003). However, fringing coral reefs found in GGBNP occupied less than 1% of the area of the Marine Park, while rocky reef is the second abundant type of habitat after soft bottom habitats at Cobourg Peninsula.

The list of species and families of fish assemblages from the three studied sites (Table 1) shows the absence of many 'typical' coral reef fish (Gladfelter *et al.*, 1980; Bell & Galzin, 1984; Choat & Bellwood, 1991; Caley, 1995; Allen, 1997; Randall *et al.*, 1997; Myers, 1999). No species from families Siganidae, Pomacentridae, Scaridae, Pempherididae, Centropomidae, Zanclidae and others were found at studied sites. It should be noted that these species are abundant on fringing coral reefs in the mouth of Port Essington (Gomelyuk, 2003). 'Typical' coral fish families represented only a small fraction of all recorded fish at studied sites (Apogonidae 3.5%, Haemulidae 2.1%, Ephippidae 2.3%, Chaetodontidae 1.4%, Serranidae 1.3%, Labridae 0.6%, Caesionidae 0.3% and Pomacanthidae only 0.2%). Only roving herbivorous Acanthuridae, a common component of coral reef assemblages comprised a substantial proportion (7.1% of all fish recorded) in fish assemblages at Site B. Carangidae, Nemipteridae, Lethrinidae, Leiognathidae and Lutjanidae dominated, representing 78.4% of all recorded fish.

None of the fish assemblages from studied sites were presented by the 'pure', single habitat-associated group—at all three sites fish communities were a mixture of fish with various habitat preferences. Reef-associated species have a larger proportion at degraded coral reef at Site B and rocky reef at Site C (Table 4). Still, fish usually associated with soft-bottom habitat made substantial fractions in fish assemblages at both Sites B and C. Conversely, fish usually associated with a coral reef environment were found in notable proportions (16.6%) on sandy bank Site A where a hard bottom habitat is absent. This bank is surrounded by vast

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Table 3. Comparison of fish assemblages at Site A (sandy bank), Site B (degraded coral reef) and Site C (rocky reef) in Port Essington. Data on average abundance (the number of fish in a video sample) of 31 of the most common fish species were used during SIMPER analysis. Cut off for low contribution: 90.0%.

	Sandy bank	Degraded coral reef			
Fish assemblages dissimilarity, δ = 88.78%					
Species	**Average abundance**	**Average abundance**	**Average dissimilarity, δ**	**δj%**	**Σ δj%**
Pantolabus radiatus	5.78	0.00	9.49	10.67	10.67
Lutjanus erythropterus	4.56	0.00	8.77	9.86	20.53
Leiognathus equulus	4.33	1.33	8.31	9.34	29.88
Pentapodus porosus	2.78	2.67	7.28	8.18	38.06
Acanthurus grammoptilus	0.00	4.80	7.19	8.09	46.14
Gnathanodon speciosus	1.56	6.27	6.73	7.57	53.71
Lethrinus ornatus	3.00	0.6	5.95	6.69	60.4
Caranx sexifasciatus	3.22	0.73	5.25	5.9	66.3
Apogon sp.	0.00	2.13	3.95	4.44	70.75
Scomberoides commersonnianus	0.67	1.67	3.25	3.66	74.4
Carnax ignobilis	0.44	0.93	2.79	3.14	77.54
Alepes apercna	0.00	2.27	2.33	2.62	80.16
Selar boops	0.00	2.20	2.06	2.31	82.47
Lutjanus carponatus	0.11	1.27	1.98	2.23	84.7
Plectorhinchus labiosum	0.11	1.13	1.88	2.11	86.82
Scolopsis monogramma	0.22	0.93	1.77	1.99	88.8
Epinephelus coioides	0.22	0.60	1.76	1.97	90.78
	Sandy bank	**Rocky reef**			
Fish assemblages dissimilarity, δ = 82.80%					
Species	**Average abundance**	**Average abundance**	**Average dissimilarity, δ**	**δj%**	**Σ δj%**
Lethrinus ornatus	3.00	2.57	9.49	11.46	11.46
Pentapodus porosus	2.78	2.79	9.34	11.28	22.74
Pantolabus radiatus	5.78	0.29	9.22	11.13	33.87
Lutjanus erythropterus	4.56	0.00	8.77	10.59	44.46
Leiognathus equulus	4.33	0.71	7.27	8.78	53.24
Lethrinus laticaudis	0.00	2.43	6.37	7.69	60.93
Caranx sexifasciatus	3.22	1.29	5.55	6.71	67.64
Scomberoides commersonnianus	0.67	0.79	4.13	4.99	72.63
Gnathanodon speciosus	1.56	1.43	2.91	3.52	76.15
Sphyraena jello	0.00	0.79	2.68	3.23	79.38
Lutjanus carponatus	0.11	0.79	2.41	2.91	82.29
Scolopsis monogramma	0.22	0.71	2.04	2.46	84.75
Apogon sp.	0.00	0.50	1.55	1.88	86.63
Acanthurus grammoptilus	0.00	0.50	1.54	1.86	88.49
Echeneis naucrates	0.78	0.21	1.52	1.83	90.32
	Degraded coral reef	**Rocky reef**			
Fish assemblages dissimilarity, δ = 86.14%					
Species	**Average abundance**	**Average abundance**	**Average dissimilarity, δ**	**δj%**	**Σ δj%**
Pentapodus porosus	2.67	2.79	10.29	11.95	11.95
Lethrinus ornatus	0.60	2.57	8.26	9.59	21.54
Acanthurus grammoptilus	4.80	0.50	7.26	8.43	29.97
Gnathanodon speciosus	6.27	1.43	6.98	8.11	38.07
Lethrinus laticaudis	0.00	2.43	6.37	7.40	45.47
Scomberoides commersonnianus	1.67	0.79	5.17	6.01	51.48
Apogon sp.	2.13	0.5	4.76	5.53	57.00
Caranx sexifasciatus	0.73	1.29	4.02	4.66	61.67
Lutjanus carponatus	1.27	0.79	3.23	3.75	65.42
Scolopsis monogramma	0.93	0.71	3.11	3.61	69.03
Sphyraena jello	1.27	0.79	3.10	3.60	72.63
Leiognathus equulus	1.33	0.71	2.78	3.22	75.86
Carnax ignobilis	0.93	0.00	2.33	2.71	78.56
Alepes apercna	2.27	0.00	2.33	2.71	81.27

Continued

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Table 3. Continued

	Degraded coral reef	Rocky reef			
Fish assemblages dissimilarity, $\delta = 86.14\%$					
Species	**Average abundance**	**Average abundance**	**Average dissimilarity, δ**	**δj%**	**Σ δj%**
Selar boops	2.20	0.00	2.06	2.39	83.65
Pantolabus radiatus	0.00	0.29	1.90	2.21	85.87
Plectorhinchus labiosum	1.13	0.07	1.89	2.20	88.07
Epinephelus coioides	0.60	0.07	1.62	1.88	89.95
Platax teira	1.73	0	1.52	1.77	91.72

δj% is percentage contribution of j-th taxa to the average Bray–Curtis dissimilarity (δ) between the groups;
Σ δj% is a cumulative contribution percentage. Species are listed in decreasing order of importance in contribution toδ.

areas of deeper soft bottom habitats and the nearest rocky reef area is 1.8 km away (Figure 1).

Ubiquitous species—fish with wide distribution and low selectivity in habitat preferences comprised a significant part in fish assemblages at all studied sites (Table 4).

At each of all three studied sites fish assemblages, while overlapping, are still distinct and significantly differ from each other. Substantial fraction of fish at all three sites (38.1% at Site A, 32.5% at Site B and 27.4% at Site C) belong to ubiquitous species. Species with broad distribution offshore and inshore waters on soft bottoms habitats and/or seagrass beds were abundant at sandy bank Site A (30.1%) and at rocky reef at Site C (31.7%). The proportion of these fish was lower at Site B (18.8%) (Table 4). Only 44.9% of all fish recorded at the edge of degraded coral reef at Site B belong to 'reef associated species'. The number of these fish at rocky reef Site C was lower (36.9%) and even lower (16.6%) at sandy bank Site A. Species found in inshore waters, in estuaries and sometimes entering river mouths were more abundant at Site A (15.2%), with only a tiny proportion of such fish at Sites B (3.8%) and C (4.0%) (Table 4).

Wide overlap between fish assemblages from Sites A and C is probably due to the high proportion of transitional species from the Carangidae family and presence at both sites of abundant species with wide distribution among a variety of habitat in the inshore zone (categories 1 and 2; Table 4).

Use of BRUVS for biodiversity surveys

Cappo *et al.* (2007) in their review of baited video system use for fish surveys highlighted an important issue: what fish species are attracted to BRUVS?

Striking differences in fish catches made by baited and unbaited fish traps (Cappo & Brown, 1996) resulted in the past in a common presumption that BRUVS are also biased towards scavenging and predatory species and exclude herbivorous and omnivorous species. However, comparison of

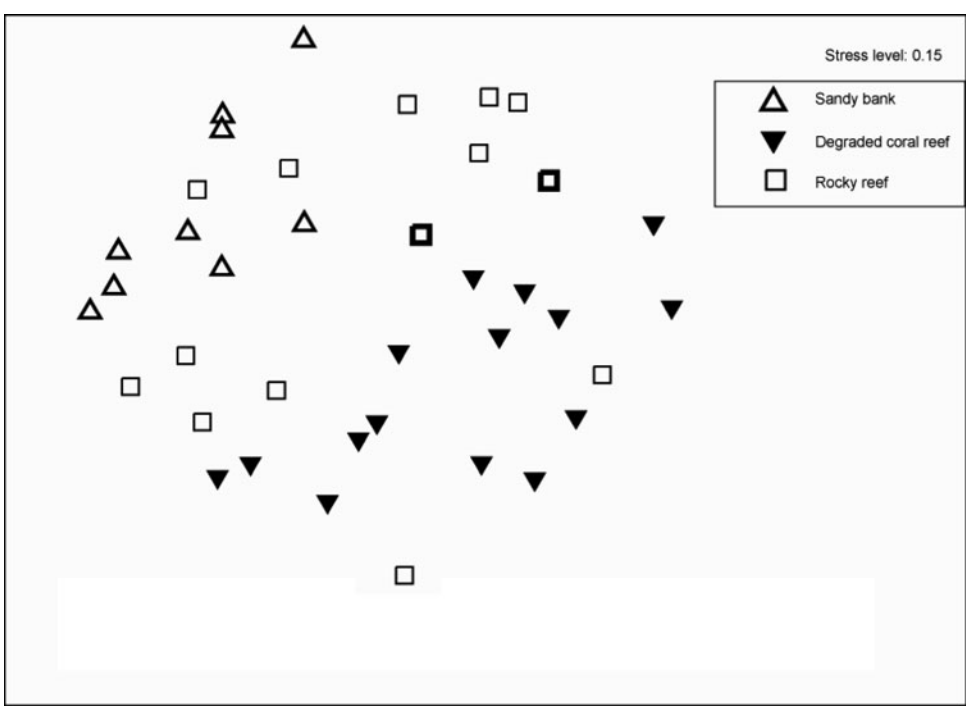

Fig. 4. Two-dimensional non-metric multidimensional scaling plots of four root-transformed abundance data in fish assemblages at three studied sites at Port Essington.

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Table 4. Fish assemblages structure according to fish habitat preferences at Site A (sandy bank), Site B (degraded coral reef) and Site C (rocky reef) in Port Essington.

Site A	%*	Site B	%*	Site C	%*
Category 1. Ubiquitous species with broad distribution found inshore and offshore; some species juveniles and sub adults in estuaries and river mouths					
Pantolabus radiatus	18.4	*Gnathanodon speciosus*	16.9	*Gnathanodon speciosus*	7.2
Caranx sexfasciatus	10.2	*Scomberoides commersonnianus*	4.5	*Caranx sexfasciatus*	6.5
Gnathanodon speciosus	4.9	*Sphyraena jello*	3.4	*Scomberoides commersonnianus*	4
Scomberoides commersonnianus	2.1	*Caranx ignobilis*	2.5	*Sphyraena jello*	3.9
Caranx ignobilis	1.4	*Caranx sexfasciatus*	2	*Lethrinus. nebulosus*	2.5
Epinephelus coioides	0.7	*Epinephelus coioides*	1.6	*Pantolabus radiatus*	1.4
Lutjanus russeli	0.4	*Scomberoides tol*	0.7	*Sphyraena obtusata*	0.7
		Lutjanus argentimaculatus	0.5	*Caranx tille*	0.4
		Carangoides chrysophrys	0.2	*Scomberoides tol*	0.4
		Remora sp.	0.2	*Epinephelus coioides*	0.4
Per cents sum	38.1	Per cents sum	32.5	Per cents sum	27.4
Category 2. Species found offshore and inshore on soft bottoms habitats and/or seagrass beds					
Lutjanus erythropterus	14.5	*Alepes* sp.	6.1	*Lethrinus ornatus*	12.9
Lethrinus ornatus	9.5	*Selar boops*	6.0	*Lethrinus laticaudis*	12.2
Remora sp.	2.5	*Plectorhinchus labiosum*	3.1	*Parachaetodon ocellatus*	2.5
Carcharinhus sp.	1.8	*Lethrinus ornatus*	1.6	*Monacanthus chinensis*	0.7
Alectis indicus	1.1	*Arius thallasinus*	0.9	*Remora* sp.	1.1
Arius thallasinus	0.7	*Alectis indicus*	0.7	*Alectis indicus*	1.1
		Upeneus vittatus	0.4	*Ulua aurochs*	0.4
				Carcharinhus sp.	0.4
				Negaprion acutidens	0.4
Per cents sum	30.1	Per cents sum	18.8	Per cents sum	31.7
Category 3. Species found mainly on coral and rocky reefs and/or on sandy areas adjacent to the reef					
Carcharhinus melanopterus	2.1	*Acanthurus grammoptilus*	13	*Pentapodus porosus*	14
Lutjanus malabaricus	1.4	*Pentapodus porosus*	7.2	*Lutjanus carponatus*	3.9
Taeniura meyeni	0.7	*Apogon* sp.	5.8	*Scolopsis monogramma*	3.6
Taeniura sp.	0.7	*Platax teira*	4.7	*Acanthurus grammoptilus*	2.5
Stegostoma fasciatum	0.4	*Lutjanus carponatus*	3.4	*Apogon* sp.	2.5
Nebris furrugineus	0.4	*Scolopsis monogramma*	2.5	*Scolopsis* sp.	2.1
Plectorhinchus labiosum	0.4	*Nebris ferrugineus*	2.2	*Chelmon marginalis*	1.4
Lutjanus carponatus	0.4	*Chelmon marginalis*	1.3	*Caesio cunning*	1.1
Pentapodus porosus	8.7	*Choerodon cyanodus*	1.1	*Lethrinus* sp.	1.1
Scolopsis monogramma	0.7	*Plectorhinchus gibbosus*	0.9	*Chaetodontoplus duboulayi*	0.7
Scolopsis sp.	0.7	*Heniochus acuminatus*	0.7	*Trachinotus blochii*	0.4
		Taeniura meyeni	0.4	*Carcharhinus melanopterus*	0.36
		Plectropomus maculatus	0.4	*Stegostoma fasciatum*	0.36
		Trachinotus blochii	0.4	*Nebris furrugineus*	0.36
		Carcharhinus melanopterus	0.18	*Taeniura meyeni*	0.36
		Triaenodon obesus	0.18	*Cephalopholis* sp.	0.36
		Aetobatus narinari	0.18	*Plectorhinchus labiosum*	0.36
		Lutianus gibbus	0.18	*Heniochus acuminatus*	0.36
		Lethrinus sp.	0.18	*Choerodon cyanodus*	0.36
				Feroxodon multistriatus	0.36
				Lagocephalus lunaris	0.36
Per cents sum	16.6	Per cents sum	44.9	Per cents sum	36.9
Category 4. Species found in inshore waters, in estuaries and sometimes entering river mouths					
Leiognathus equulus	13.7	*Leiognathus equulus*	3.6	*Leiognathus equulus*	3.6
Elops hawaiensis	0.7	*Rhinobatus typus*	0.2	*Pastinachus sephen*	0.4
Rhinobatus typus	0.4				
Pastinachus sephen	0.4				
Per cents sum	15.2	Per cents sum	3.8	Per cents sum	4.0

%*, per cents of all fish individuals recorded at the site.

baited and unbaited remote video stations showed that baited video appears to be the best technique to obtain recordings of a large number of species and individuals (Willis & Babcock, 2000; Watson *et al.*, 2005). Use of bait actually increased the ability to discriminate fish assemblages in distinctive benthic habitats in tropical and temperate Australia due to the increased number of individuals and species sampled at the baited stations (Watson *et al.*, 2005; Cappo *et al.*, 2007; Harvey *et al.*, 2007). Baited video consistently sampled more individuals and species of the fish recognized as herbivores, feeders on invertebrates and algae than unbaited stations (Cappo *et al.*, 2007) without decreasing the abundances of

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herbivorous or omnivorous fish (Harvey *et al.*, 2007). Omnivore and planktivore species were among regular visitors at the bait during the BRUVS study in the western Mediterranean (Stobart *et al.*, 2007). This can be resulted from a 'sheep effect' when a few individuals that approached the bait attracted others to do so too (Watson *et al.*, 2005), from aggregation response where other fish activity is detected (Manteifel, 1970; Priede & Merret, 1996, 1998). That is why fish ignoring bait such as herbivorous *Acanthurus grammoptilus* and *Leiognathus equulus*, a generalized carnivore (Sommer *et al.*, 1996), have both a high rank in the recorded species list (Table 1). Observations made in this study show that all sharks and teleost fish from families Lutjanidae, Nemipteridae, Haemulidae, Lethrinidae and Ephippidae were feeding from a bait canister. It is important to add, that piscivorous and carnivorous fish from families Carangidae, the most abundant group of recorded fish, were never observed taking bait from BRUVS. These fish and large piscivorous *Sphyraena jello* usually row through an area or circling around a group of fish feeding from the bait canister without paying any interest to the bait. Piscivorous and carnivorous Serranidae also seemed to be attracted by fish movement around the bait canister but have never been observed to take the bait. Tape interrogations revealed that only 37.5% of all 1115 recorded fish individuals (Table 1) were approaching the bait canister and feeding. Therefore, it may be concluded that the bait used during survey concentrates existing fish assemblages in the BRUVS area, rather than altering 'natural' assemblages by selective attraction of some species to the field of view of the camera. This study supports the idea (Cappo *et al.*, 2007) that the BRUVS technique is an effective method for fish assessment. It can be successfully used in areas where other visual methods cannot be applied—even if water clarity is very poor and unsuitable for diving visual and video surveys. Because of the non-impact nature of visual surveys they can be used in long-term environment monitoring of fish assemblages at reference sites.

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